

CHIHABEDINNE BOURAS

Electrical Engineering & Embedded Systems Student

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PROFILE

Third-year student in Electrical Engineering and Industrial Computing (BUT GEII), Electronics and Embedded Systems track, at IUT de Troyes. Focused on the design and implementation of embedded systems (VHDL/FPGA, C/C++, industrial automation). Aiming to pursue graduate-level research in embedded systems through admission to a French école normale supérieure, with the long-term goal of contributing to research in electronics and embedded systems. Rigorous and autonomous, I enjoy carrying a project through from functional analysis to bench validation.

SKILLS

Software & tools

- Qt Creator — C++ application development and debugging, GUI design
- Quartus — digital logic design, simulation and synthesis in VHDL for FPGA
- Unity Pro — PLC programming (Ladder, Grafcet/SFC)
- MATLAB, Arduino IDE, Tinkercad, Schemaplic

Programming & hardware description languages

- C / C++ — performance-oriented development, memory management, OOP
- VHDL — digital circuit design and simulation, FPGA/CPLD modules for embedded systems
- Ladder / SFC / ST — industrial control-command systems, logic sequencing
- Python, Arduino C

LANGUAGES

- French — Native
- English — Intermediate (B1), effective in professional and everyday contexts
- Spanish — Elementary (A2), good written and spoken comprehension

EDUCATION

BUT (Bachelor Universitaire de Technologie) — Electrical Engineering and Industrial Computing (GEII), Electronics and Embedded Systems track *IUT de Troyes, 2024–2027*

French Baccalauréat, Science track, with honours *Lycée Camille Claudel, Troyes, 2023–2024*

SELECTED PROJECTS

Autonomous vehicle — software subsystem

- Designed the battery supervision module and a Qt/C++ client interface communicating over TCP for real-time monitoring and control.

Tennis-playing robot

- Developed a robot capable of autonomously locating and returning a ball on a tennis court.

Loudspeaker impedance measurement bench

- Built an analog (Howland, Rauch, MPY634) and digital (SPI ADC/DAC, FFT on Arduino) acquisition chain to characterise loudspeaker impedance and convolution behaviour, validated against a MATLAB model.

Motor control on FPGA/CPLD

- Implemented VHDL logic on a DE10-Lite CPLD for DC motor control: PWM generation, incremental encoder reading, 7-segment display.

Autonomous obstacle-avoidance robot (Escape Bot)

- Built obstacle avoidance and beacon detection on a Raspberry Pi Pico, integrating a Pixy camera, IMU and LiDAR sensors; designed the battery supervision PCB.

Temperature regulation system

- Designed and built a modular temperature measurement and control system across 7 stripboard-mounted function cards.

WORK EXPERIENCE

Logistics Operator — GSF Ariane *Jun–Aug 2025*

- Reception, scanning and sorting of packages in a high-volume sorting facility; maintained a safe, organised work environment.

Event Security Agent — McArthur Glen, Troyes *2023–2024*

- Coordinated schedules and team movements; managed unexpected situations to ensure the safety of participants and venues.

Internship — SNCF, Gare du Nord, Paris *2020–2021*

- Assisted an employee in daily operations to understand core business processes; built organisational and communication skills.